MIL-E-1/1359A
24 February 1971
SUPERSEDING
MIL-E-1/1359(SigC)
1 August 1960

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 6028

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: Pentode, miniature, rf, sharp cutoff

Outline	---	5-1 (EIA)
Base	---	E7-1
Envelope	---	T5-1/2
Cathode	---	Coated unipotential

Base connections:

		Ⓐ						
Pin No.	---	1	2	3	4	5	6	7
Element	---	g1	g3, k	h	h	a	g2	g3, k

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	Eb	Ec1	Ec2	Rk	Ik	Pp	Pg2	Ⓐ Ehk	Ⓐ Alt
Unit:	V	Vdc	Vdc	Vdc	Ohms	mAdc	W	W	v	ft
Maximum:	22	200	---	155	---	20	1.85	0.55	75	(See note 1)
Minimum:	18	---	---	---	---	---	---	---	---	
TEST CONDITIONS:	20	120	0	120	200	---	---	---	---	---

GENERAL:

Qualification - Required

Ⓐ denotes changes

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1							
1266	Total grid current	See notes 2 and 3	0.65	II	Ic1	0	-0.1	µAdc
1256	Electrode current (1) (anode)		0.65	II	Ib	5.0	9.0	mAdc
1256	Electrode current (2) (anode)	Ec1 = -10 Vdc	0.65	II	Ib	0	200	µAdc
1256	Electrode current (screen)		0.65	II	Ic2	1.5	3.0	mAdc
1306	Transconductance (1)	See note 2	0.65	II	Sm	3,900	6,000	µmhos
1306	Transconductance (2)	Ef = 18 V	0.65	II	ΔSm	---	11	%
1201	Short and discontinuity detection		0.4	II	---	---	---	---
	Quality conformance inspection, part 2							
(A) 1031	Low-frequency vibration	Rp = 10,000 ohms	---	---	Ep	---	200	mVac
1301	Heater current		2.5	I	If	45	55	mA
1336	Heater-cathode leakage		2.5	I	Ihk	0	30	µAdc
(A) ---	Grid drive	See note 4	---	---	---	---	---	---
(A) 1331	Direct-interelectrode capacitance	Shield No. 316	---	---	Cgp	---	0.02	pF
					Cin	3.4	4.4	pF
					Cout	2.45	3.25	pF
1121	Base strain		---	---	---	---	---	---
2126	Glass strain		2.5	I	---	---	---	---
(A) 1211	Insulation of electrodes		---	---	---	---	---	---
(A) 1105	Permanence of marking		---	---	---	---	---	---

METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSP LEVEL OR CODE	ALLOWABLE DEFECTIVES PER CHARACTERISTIC 1 ST SAMPLE	ALLOWABLE DEFECTIVES PER CHARACTERISTIC COMBINED SAMPLES	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 3									
---	Life test	Group A; Ehk = 75 V	---	---	---	---	---	---	---	---
---	Life-test end point (500 hours)	Transconductance (1)	---	---	1	3	Sm	3,200	---	µmhos

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NOTES:

1. See "Reduced pressure (altitude) rating", and altitude, maximum peak voltage in the basic document.
2. This test to be performed at the conclusion of the holding period.
3. Approved alternate test. The insertion of a 2.7 - Meg resistor in the grid circuit shall not change the anode current more than 0.6 mAdc.
4. This test shall be made with the control grid, screen grid, and anode connected together, and in a circuit such that these elements are driven not less than 9 volts positive peak 60 Hz ac, in respect to the cathode. After operation for 10 minutes under these conditions the tube shall:

   (a) Meet the minimum anode current (1) requirements.
   (b) Meet the minimum transconductance (1) requirements.
   (c) Not exceed the maximum transconductance (2) requirements: ΔSm = 16 percent maximum.

   Applicable to lot acceptance, the sample tested shall be considered to have passed this test if it has met the above requirements; however, tubes found to exceed the maximum anode current (1) or transconductance (1) limits shall not be delivered on a contract or order.

Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Army - EL
Air Force - 11, 17, 80
DSA - ES

User activities:
Army - WC
Navy - AS, OS, MC, CG, SH
Air Force - 19

Preparing activity: Army - EL

Agent: DSA - ES

(Project 5960-2468)

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Department of the Army
Headquarters
U.S.Army Electronics Command
Fort Monmouth, New Jersey 07703

POSTAGE AND FEES PAID

OFFICIAL BUSINESS

Commanding General
U.S.Army Electronics Command
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Fort Monmouth, New Jersey 07703

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION
MIL-E-1/1359A, Electron Tube, Receiving Type 6028

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
   ☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

ESC-FM 1068-68

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